

WORK SHEET - 10

PARABOLA

[SINGLE CORRECT CHOICE TYPE]

Q.1 to Q.4 has four choices (A), (B), (C), (D) out of which **ONLY ONE** is correct.

- Q.1 Let a ray of light is travelling along the line $4x + 3y - 2k = 0$ and strikes a parabola at $(1, p)$. After reflection it moves along the line $3x - 4y + k = 0$. If the directrix of the parabola is the line $3x - 4y + 15 = 0$ then the latus rectum of the parabola is
(A) 4 (B) 6 (C) 8 (D) 16
- Q.2 A chord PQ is a normal to the parabola $y^2 = 4ax$ at P and subtends a right angle at the vertex. If $SQ = \lambda SP$ where S is the focus then the value of λ , is
(A) 1 (B) 2 (C) 3 (D) 4
- Q.3 A ray of light moving parallel to the x-axis get reflected from a parabolic mirror whose equation is $(y - 2)^2 = 4(x + 1)$. After reflection, the ray must pass through the point
(A) (0, 2) (B) (2, 0) (C) (0, -2) (D) (-1, 2)
- Q.4 If the line $y - \sqrt{3}x + 3 = 0$ cut the parabola $y^2 = x + 2$ at A and B, then the value of $PA \cdot PB$ is [where $P = (\sqrt{3}, 0)$]
(A) $\frac{4(2 + \sqrt{3})}{3}$ (B) $\frac{4(2 + \sqrt{3})}{\sqrt{3}}$ (C) $\frac{4(2 - \sqrt{3})}{3}$ (D) $\frac{4(2 - \sqrt{3})}{\sqrt{3}}$

[PARAGRAPH TYPE]

Q.5 to Q.14 has four choices (A), (B), (C), (D) out of which **ONLY ONE** is correct.

Paragraph for question nos. 5 & 6

Normals are drawn from $P(4, 1)$ to the parabola $y^2 = 4x$. The tangents at the feet of normals to the parabola $y^2 = 4x$ form a triangle ABC.

- Q.5 The centroid of triangle ABC is equal to
(A) $\left(-\frac{4}{3}, 0\right)$ (B) $\left(\frac{-1}{3}, \frac{1}{3}\right)$ (C) $\left(\frac{1}{3}, \frac{-1}{3}\right)$ (D) $\left(\frac{-2}{3}, 0\right)$
- Q.6 The area of triangle ABC is equal to
(A) $\frac{\sqrt{5}}{4}$ (B) $\frac{\sqrt{5}}{8}$ (C) $\frac{\sqrt{5}}{2}$ (D) $\frac{\sqrt{5}}{16}$

Paragraph for question nos. 7 & 8

Consider a circle $S : (x + 2)^2 + y^2 = 1$ and a parabola $P : y^2 = 4x$

- Q.7 If n is number of common tangents to both the curves and $m_1, m_2, \dots, m_n$ are the slopes of these tangents then $\sum_{i=1}^n m_i^2$ equals
(A) 5/3 (B) 10/3 (C) 20/3 (D) 10
- Q.8 If pair of tangents are drawn from each and every point on the circle S to the parabola P, then maximum perpendicular distance from the origin to the corresponding chord of contact is
(A) 3 (B) 4 (C) 5 (D) 6

Paragraph for question nos. 9 to 11

Consider the parabola $x^2 = 4y$ and circle $C : x^2 + (y - 5)^2 = r^2$ ($r > 0$). Given that the circle C touches the parabola at the points P and Q .

- Q.9 Radius of the circle C is
(A) 2 (B) 3 (C) 4 (D) 5
- Q.10 The distance between chord PQ and directrix of the parabola is
(A) $5/3$ (B) 3 (C) 4 (D) $4/3$
- Q.11 The equation of the circle which passes through the vertex of the parabola $x^2 = 4y$ and touches it at the point $M(-4, 4)$, is
(A) $x^2 + y^2 - 10x - 18y = 0$ (B) $x^2 + y^2 - 8x - 16y = 0$
(C) $x^2 + y^2 + 8x = 0$ (D) $x^2 + y^2 - 8y = 0$

Paragraph for question nos. 12 to 14

Consider the following lines $L_1 : x = y + 1$, $L_2 : x + y = 5$ and $L_3 : y = 4$. Let L_1 is axis to a parabola, L_2 is tangent at the vertex to this parabola and L_3 is another tangent to this parabola at some point P . Let C be the circle circumscribing the triangle formed by tangent and normal at point P and axis of parabola. The tangent and normal at the extremities of latus rectum of this parabola forms a quadrilateral $ABCD$.

- Q.12 The equation of circle C is
(A) $x^2 + y^2 - 2x = 31$ (B) $x^2 + y^2 - 2y = 31$ (C) $x^2 + y^2 - 2x - 2y = 31$ (D) $x^2 + y^2 + 2x + 2y = 31$
- Q.13 Which of the following parabola is equal to the given parabola?
(A) $y^2 = 16\sqrt{2}x$ (B) $x^2 = -4\sqrt{2}y$ (C) $y^2 = -\sqrt{2}x$ (D) $y^2 = 8\sqrt{2}x$
- Q.14 The area of quadrilateral $ABCD$ is equal to
(A) 8 (B) 16 (C) 32 (D) 64

[MULTIPLE CORRECT CHOICE TYPE]

Q.15 to Q.18 has four choices (A), (B), (C), (D) out of which **ONE OR MORE** may be correct.

- Q.15 The common tangents of the circle $(x + 1)^2 + y^2 = 1$ and the parabola $y^2 = 4x$ is(are)
(A) $x = 0$ (B) $y = \frac{1}{\sqrt{3}}x + \sqrt{3}$ (C) $y = \frac{-1}{\sqrt{3}}x - \sqrt{3}$ (D) $y = x$
- Q.16 Suppose two tangents drawn to a parabola at points A and B on it are $x + y + 2 = 0$ and $x - y + 4 = 0$. While the normals drawn at A and B meet at $C(1, 2)$ lying on its axis, then
(A) Equation of axis of parabola is $x - 4y + 7 = 0$.
(B) Length of Latus rectum of parabola is $\sqrt{17}$.
(C) Equation of axis of parabola is $x + 4y - 9 = 0$
(D) Length of Latus rectum of parabola is $\sqrt{15}$.
- Q.17 Let PQ be a chord of parabola $y^2 = 4x$. A circle drawn with PQ as a diameter passes through the vertex V of the parabola. If area $(\Delta PVQ) = 20$, then the coordinates of P are
(A) $(16, 8)$ (B) $(16, -8)$ (C) $(-16, 8)$ (D) $(-16, -8)$
- Q.18 The circle $x^2 + y^2 - 4x = 6$ and the parabola $y^2 = 9x$ intersect at P and Q in the first and the fourth quadrants respectively. Tangents to the circle at P and Q intersect the x -axis at R and tangents to the parabola at P and Q intersect the x -axis at S . Then
(A) length of PQ equals 8.
(B) radius of circle described on RS as diameter equals $7/2$.
(C) $(1, -1)$ is a point on the common chord of circle and parabola.
(D) area of triangle PRS equals $27/2$.

[INTEGER TYPE]

Q.19 & Q.20 are "Integer Type" questions. (The answer to each of the questions are upto 4 digits)

- Q.19 Chord of the curve $3x^2 - y^2 - 2x + 4y = 0$, which subtend a right angle at the origin, always passes through vertex of a concave up parabola whose axis is parallel to y-axis and length of latus rectum is 8. If extremity of latus rectum of parabola is $L_1(\alpha, \beta)$, $L_2(\gamma, \delta)$, then find $(\alpha + \beta + \gamma + \delta)$.
- Q.20 Consider a parabola $y^2 = -16x$ and $P(x, y)$ be any point lying on it. If maximum and minimum value of $z = \frac{2(y-1)}{2x-1}$ are M and m respectively, then find $|M + m|$.

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ANSWER KEY

Q.1	A	Q.2	C	Q.3	A	Q.4	A	Q.5	D
Q.6	C	Q.7	B	Q.8	A	Q.9	C	Q.10	C
Q.11	B	Q.12	A	Q.13	D	Q.14	D	Q.15	ABC
Q.16	AB	Q.17	AB	Q.18	BC	Q.19	[2]	Q.20	[2]

